

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI
DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY



MASTER THESIS

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Hanoi, <Month – yyyy>

To whom it may concern,

I, (supervisor's name), certify that the
thesis/ internship report of Mr/Ms.(student's
name) is qualified to be presented in the Internship M1 Jury 2025-2026.

Hanoi, <date>

Supervisor's signature

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ACKNOWLEDGEMENTS

Example of acknowledgements

“I would like to express my gratitude to all those who gave me possibility to complete this thesis in specific as well as academic supporting in general that I received during the time of my studying at the University of Science and Technology of Hanoi.

Foremost, I would like to send my honest thanks to my supervisor...”

LIST OF ABBREVIATIONS

Example of list of abbreviations

BP	Biotechnology- Pharmacology
CMOS	Complementary Metal-Oxide-Semiconductor
CMRR	Common-Mode Rejection Ratio
ICT	Information and Communication Technology
NMC	Nested Miller Compensation
GB	Gain Bandwidth
USTH	University of Science and Technology of Hanoi

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ABSTRACT

(No more than 250 words, in English + 6 key words)

Example of abstract

We present a fully-integrated compact (1.1 mm²) target device for Near Field Communication (NFC). The design of the key analog part of the tag IC is presented, which includes a robust demodulator for 10% ASK envelope detection, a high quality random number generator, an adaptive RF limiter, and a low power clock generator. A 128 bit advanced encryption standard (AES) with new cyclic key generation is used for secure data encryption and decryption. An on-chip 4Kb EEPROM is used to support the AES operation. The tag chip is fabricated in a 1-poly 6-metal low-power (LP) 0.18 μm CMOS process with a CoSi₂-Schottky diode and EEPROM process.

Key words: Near Field Communication, ASK, AES, low power, Schottky diode, EEPROM process.

I/ INTRODUCTION

(Global context, literature review, main questions and objectives including the presentation of the problem with the scientific background, 2-3 pages)

II/ OBJECTIVES

Scientific objective of the internship: submit your goal and summarize the strategy in 2-3 concise sentences

III/ MATERIALS AND METHODS

(Presenting in this section the materials and the methods you have used. Methods have to be presented not in the form of a bench protocol but rather in a style suitable for publication (a paragraph of complete sentences) and include all information necessary for a third person to reproduce your experiments (in detail but condensed))

IV/ RESULTS AND DISCUSSION

1. **Results:** Description of the student's own research, procedures and results. Interpretation of the student's results; conclusions and review of the results comparison with other research; critical assessment of the student's own work; limitation of the study).
2. **Discussion:** in this section you have to do a critical analysis and discuss the results you obtained in regards to the data already available in the scientific literature. In case of failure you have to analyze the reasons and propose an alternative strategy. In case of success you have to state the essential new information your work has given to solve the scientific problem and about the immediate perspective that it opens.

Example of table to present your results (always choose between table or figure)

Table 1: Topographic parameters measured for the 15 gully heads in 2009

GULLY NUMBER	Drain.A (m²)	DAslope (%)	LHS (%)	SPI	GL (m)	GHA (m²)	GVol (m³)	RL (m y⁻¹)	Retreat.A (m² y⁻¹)	V (m³ y⁻¹)
1	41258	28.4	17.3	3.78	120	4844	7673	0.67	68.6	218
2	85597	29.4	30.1	5.39	225	11695	28096	0.33	202.0	727
3	8490	12.1	10.2	3.05	98	2258	2090	1.33	56.3	150
4	105150	30.9	12.2	3.83	218	8101	53650	0.33	56.6	358
5	26467	18.8	6.4	2.26	143	4175	22030	0.33	14.6	69
6	13054	18.8	12.9	3.56	68	875	1998	0.33	4.0	11
7	17412	26.7	13.9	3.82	69	809	1459	0.67	19.0	64
8	14350	17.7	8.8	4.78	71	1393	1875	1.33	38.6	122
9	103776	24.2	10.0	1.89	240	12008	79928	1.33	163.6	1303
10	3701	12.9	8.7	1.69	65	1795	4235	0.67	52.6	266
11	2790	9.1	5.4	0.79	57	555	423	0.33	15.0	17
12	57941	30.4	9.1	4.14	63	1196	4429	1.00	33.0	157
13	38924	28.2	7.1	3.17	132	3585	23642	1.67	113.0	897
14	9240	36.4	15.7	3.30	73	1292	2995	0.67	43.6	156
15	3039	10.5	12.2	1.87	64	911	1922	0.67	35.0	123
mean	35412	22.3	12.0	3.20	113	3699	15763	0.77	61.0	309
SD	36336	8.6	6.0	1.05	64	3881	23140	0.44	56.5	374

Table 2: Simulation setup

Symbol	Value	Description
N	10,000	Number of nodes in the network
S	1000x1000(m ²)	Network deployment area
r	40m	The communication range
M	200 keys	The memory size of nodes for storing key materials.
σ	50m	The standard deviation in Gaussian distribution.

Example of figure to present your results:

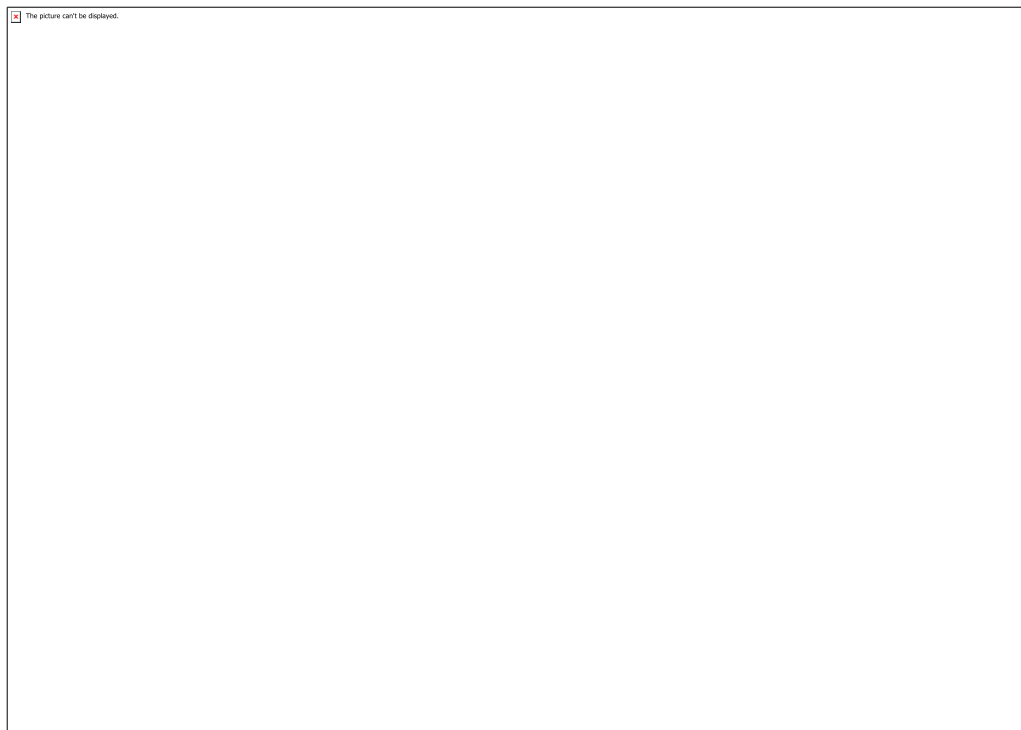


Figure 1: Hexagon group-based deployment model

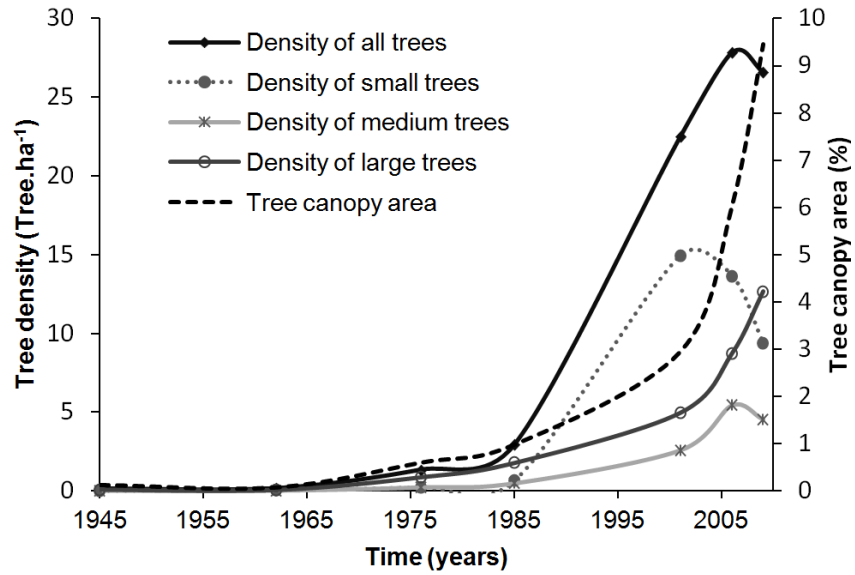


Figure 2: Smoothed tree density and tree canopy area in the catchment from 1945 to 2009

Smoothed tree density and tree canopy area in the catchment from 1945 to 2009. Small trees (canopy area < 1 m²), medium-sized trees (1 m² < canopy area < 15 m²), large trees (canopy area > 15 m²).

Note: Never start your result section with a Fig or Table only. An introduction paragraph should be added.

V/ CONCLUSION & PERSPECTIVE

The conclusion recapitulates the main achievements of your work and the main perspectives it opens.

REFERENCES

Please ensure that every reference cited in the text is also present in the references list (and vice versa);

References style:

1. Text: Indicate references by number(s) in square brackets in line with the text. The actual authors can be referred to, but the reference number(s) must always be given.
 - *Example:* '..... as demonstrated [3,6]. Barnaby and Jones [8] obtained a different result'.
2. References list: Number the references (numbers in square brackets) in the list in the order in which they appear in the text.
Reference formatting: author(s) name(s), journal title/ book title, chapter title/article title, year of publication, volume number/book chapter (issue), and the article number or pagination must be present. Use of DOI is highly encouraged. For articles/books with more than 10 authors, list the first 10 followed by “et al.”
 - *Examples:*
 - Reference to a journal publication:
[1] J. van der Geer, J.A.J. Hanraads, R.A. Lupton, The art of writing a scientific article, J. Sci. Commun. 2010, 163 (2), pp. 51–59.
<https://doi.org/10.1016/j.Sc.2010.00372>.
Reference to a journal publication with an article number:
[2] Van der Geer, J., Hanraads, J.A.J., Lupton, R.A., 2018. The art of writing a scientific article. Heliyon. 19, e00205.
<https://doi.org/10.1016/j.heliyon.2018.e00205>.
 - Reference to a book:
[3] W. Strunk Jr., E.B. White, The Elements of Style, fourth ed., Longman, New York, 2000.
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[4] G.R. Mettam, L.B. Adams, How to prepare an electronic version of your article, in: B.S. Jones, R.Z. Smith (Eds.), Introduction to the Electronic Age, E-Publishing Inc., New York, 2009, pp. 281–304.
 - Reference to a website:
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<http://www.cancerresearchuk.org/aboutcancer/statistics/cancerstatsreport/> , 2003 (accessed 13 March 2003).
 - Reference to a dataset:
[dataset] [6] M. Oguro, S. Imahiro, S. Saito, T. Nakashizuka, Mortality data for Japanese oak wilt disease and surrounding forest compositions, Mendeley Data, v1, 2015. <https://doi.org/10.17632/xwj98nb39r.1>.

APPENDICES

APPENDIX 1

APPENDIX 2